

Sepehr Kahvand

M.Eng in Mechanical Engineering | Mechanical Designer | CSWA | GD&T & Tolerance Design
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SUMMARY

Mechanical Engineering Professional with 9 years of experience in mechanical design, production tooling, and manufacturing support in automotive and industrial environments. SolidWorks CSWA-certified with advanced capability in 3D modeling, assemblies, and engineering documentation. Strong in GD&T (ASME Y14.5) and DFM, producing accurate models, detailed drawings, and BOMs for custom assemblies, jigs, fixtures, and tooling. Proven record of improving production efficiency and reducing costs through process optimization. Knowledgeable in materials, manufacturing processes, and ISO 9001 / IATF 16949 quality systems.

RELATED PROFESSIONAL EXPERIENCES

Linamar Corporation (*Comtech Mfg.*)

❖ **Computer Numerical Control Machine Operator** – *Permanent Full-time*
Aug 2025 – Present

- Operate **CNC lathe machines** to manufacture transmission cover plates in a **fast-paced, high-volume production environment**
- Perform **tool insert changes, machine adjustments, and offsets** to ensure parts meet nominal dimensions and quality specifications
- Inspect machined parts every 2 hours using **Mueller gauge, height gauge, and Go/No-Go gauges**, completing **Operator Data Sheets** for QA documentation
- Identify and report **non-conforming products**, tag defective parts, and halt production until approval from supervisors or team leaders
- **Improved production output by 47% by analyzing the automation workflow and collaborating with the program manager and automation supervisor to optimize the logic of the automated system**

University of Guelph

❖ **Teaching Assistant of the course “ENGG*4490 Sampled Data Control Design”** – *Contract Part-time*
Jan 2025 – Apr 2025

- Facilitated lab experiments on Coupled-Tanks and Magnetic Levitation systems, supporting applied learning in digital control systems
- Evaluated lab reports and provided constructive technical feedback to enhance students' analytical skills and engineering communication
- Assisted students in implementing digital controllers using MATLAB/Simulink, helping them bridge theoretical concepts with real-time system behavior
- **Improved student lab performance and comprehension by providing tailored guidance and practical troubleshooting, contributing to a 10% increase in average lab grades**

Faraz Sanat Engineering Co.

❖ **Mechanical Designer** – *Permanent Full-time*
Apr 2020 – Apr 2024

- Designed mechanical subassemblies, custom jigs, dies, fixtures and production tooling using **SolidWorks**
- Performed tolerance stack-ups and GD&T analysis to ensure manufacturability and assembly reliability
- Used ANSYS for basic stress and deformation analysis on tooling components to validate design robustness
- Generated and maintained accurate **bills of materials (BOMs)** and drawing packages for internal and supplier use
- **Led tooling and fixture design improvements through detailed 3D modeling and GD&T application, collaborating with engineers and production teams to achieve 2.7% production line efficiency increase, reduced assembly time, and lower manufacturing errors—delivering estimated annual savings of \$185,000**

Niroo Motor Group

❖ **Quality Assurance Specialist / Quality Control Inspector – Motorcycle Assembly Division – Permanent Full-time**
Jan 2015 – Mar 2020

- Led investigations into assembly line defects such as misalignment, paint damage, wiring errors, and loose fasteners, using root cause analysis tools (5 Whys, Fishbone)
- Issued and tracked Corrective and Preventive Actions (CAPA) for recurring issues, collaborating with assembly, logistics, and supplier teams
- Supported compliance with ISO 9001 and IATF 16949 by maintaining updated records, calibration logs, and audit-ready documentation
- Reported key quality metrics (defect rate per station, rework frequency, customer complaint trends) to QA Manager with recommendations for long-term improvements
- **Reduced rework and defect rates on the final assembly line by over 20% within two years through process optimization, better operator training, and systematic issue tracking**

EDUCATION

- Master of Engineering: Mechanical Engineering May 2024 – Apr 2025
University of Guelph, Canada | CGPA: 92/100 | **Graduated with Top Academic Standing**
- Bachelor of Science: Mechanical Engineering (Solid Mechanics and Design)
University of Guilan, Iran

CERTIFICATIONS

- **Certified SOLIDWORKS Design Associate**
Dassault Systèmes | Completed: April 12, 2026
- **Geometric Dimensioning & Tolerancing: Advanced concepts**
Udemy | Completed: April 5, 2026
- **Worker Health & Safety Awareness – Ontario Regulation 297/13**
University of Guelph | Completed: January 9, 2025
- **WHMIS 2024 – Workplace Hazardous Materials Information System**
University of Guelph | Completed: June 26, 2024
- **Advanced Python Programming and Object-Oriented Thinking Course**
Quera College | Completed: December 2022
- **Internal Quality Auditor – ISO 9001:2015 (Manufacturing Focus)**
GLIS Training Solutions | Completed: 2020

SKILLS

Technical Skills:

- | | | | |
|-----------------------------------|-----------------------|---------------------|----------------------------|
| • Computer-Aided Design (CAD) | • SolidWorks | • AutoCAD | • FMEA/FMECA |
| • BOM Documentation | • GD&T (ASME Y14.5) | • Tolerance Design | • Test Plan Development |
| • Autodesk Inventor | • Reliability Testing | • CNC lathe machine | • MATLAB/Simulink |
| • ISO 9001:2015 IATF 16949:2016 | • Python | • Microsoft Excel | • FEA (ANSYS & SolidWorks) |

Professional Competencies:

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|-------------------|-----------------------|-----------------------|-------------------|
| • Teamwork | • Communication | • Adaptability | • Time Management |
| • Problem Solving | • Analytical Thinking | • Attention to Detail | |

REFERENCES

- **References available upon request.**